Abdelwakil Khamsi

Mechanical and Robotics Engineer · Actuation, Sensing and Control of Robotic Hardware
Bournemouth, UK +44 7350 212858 abdelwakil55khamsi@gmail.com LinkedIn ↗ Portfolio ↗

PROFILE

MSc Mechanical Engineering with Distinction. Built a planar bipedal robot and its ROS 2 software architecture: URDF and Pinocchio dynamics, parametric Bézier gaits, and CasADi trajectory optimisation that minimises energy and landing impact and outputs feedforward torques for the joint control loops. Hardware paper in preparation.

EDUCATION

Bournemouth University, England Sep 2025 – Sep 2026
Master of Science (MSc) in Mechanical Engineering Distinction

- **Dissertation:** "Design, Manufacture and Model-Based Control of a Planar Bipedal Robot", grant-funded by Bournemouth University.
- **Modules:** Advanced Structural Mechanics, Failure Analysis and Prevention, Robotic Control Design, Model-Based Engineering.

Polytechnic School of Agadir, Morocco Sep 2021 – Jul 2023
Master of Science (MSc) in Mechanical Engineering: Maintenance and Quality

- **Thesis:** "Improving the Availability of the PS III Sulphuric Acid Line through the Implementation of a Shell Replacement Procedure", carried out at OCP Group.
- **Modules:** Robotics, industrial maintenance, computerised maintenance management (CMMS), quality management.

Higher Institute of Maritime Fisheries, Morocco Sep 2018 – Jul 2021
Bachelor of Science (BSc) in Maritime Mechanical Engineering

- **Thesis:** "Design and Implementation of a PID Temperature Control System for a Sardine Canning Cooking Installation".

TECHNICAL SKILLS

1. **Robotics & Control:** ROS 2 (ros2_control, RViz), Gazebo, multi-axis trajectory control, PID control, inverse kinematics, inverse dynamics (Pinocchio), Bézier trajectory generation.
2. **Optimisation & Modelling:** CasADi and IPOPT trajectory optimisation, Lagrangian dynamics, MATLAB, Simulink and Simscape Multibody.
3. **Sensing & Embedded:** ODrive brushless motor control, CAN bus, RS-485 absolute encoders, load-cell force sensing and calibration (HX711), Ubuntu Linux.
4. **Programming:** Python, C++, MATLAB.
5. **CAD & Simulation:** CATIA V5/V6, SolidWorks, Fusion 360 (incl. generative design), Autodesk Inventor, ANSYS Mechanical, Abaqus.
6. **Manufacturing:** FDM 3D printing, machining and modification of steel parts, conventional and CNC machining, rapid prototyping, GD&T, tolerance stack-up, DFM.
7. **Validation & Reliability:** design verification and validation testing, failure analysis, FMEA, FEA, root cause analysis, life cycle analysis, CMMS (Optimaint, Oracle).

RESEARCH & WORK EXPERIENCE

1. Planar Bipedal Robot: Design, Actuation, Sensing and Trajectory-Optimised Control Mar 2026 – Present
Researcher, grant-funded MSc project continued after graduation, Bournemouth University | Supervisor: Dr Behnam Dadashzadeh

- Designed and built a five-link, four-actuator planar biped on a boom-constrained walking rig, extending a single-leg design from Prof. Alexander Badri-Spröwitz's group (Mo et al., Scientific Reports, 2023) into a complete robot with eleven documented design changes. Structure 3D-printed; drive parts sourced in steel and modified in-house after metal printing and EDM proved impractical.
- Integrated four T-Motor MN7005 brushless actuators with ODrive S1 drives over a 1 Mbps CAN bus, with absolute RS-485 encoders and load-cell force sensing, controlled from ROS 2 Jazzy on a Linux PC, so all four joints are commanded and logged in sync.
- Built the ROS 2 software stack from scratch: URDF with parallelogram-linkage constraints, a ros2_control interface shared by Gazebo and the hardware, a Bézier gait generator with its own design GUI, and Pinocchio inverse dynamics.
- Characterised the robot on hardware: compliant position gains gave about 0.6 Hz of tracking bandwidth against the 5–10 Hz a swing phase needs, which made the case for model-based torque feedforward. Walking trials isolated five specific mechanical faults, each with a planned fix.

- Generated walking gaits by trajectory optimisation on the robot's full dynamic model: formulated a constrained optimal-control problem with soft touchdown, smooth weight transfer, friction and motor limits, transcribed it in CasADi and solved it with IPOPT. Verified the model against Pinocchio, validated the gaits in simulation and ran first trials on the robot, where foot slip is now the limiting factor.

2. Design Release Engineer, Segula Technologies Nov 2023 – Aug 2024

Client: Stellantis | Hydrogen fuel cell system for a heavy-duty truck

- Contributed to the design release of the fuel cell system, working with engineering teams in the US, Germany, France and Romania.
- Supported removal of the turbine bypass, cutting cost by €32.50 per part and simplifying the system without compromising durability.
- Resolved supplier issues on fuse integration, temperature sensors, engine mounts and subsystem mounting points.
- Validated the "eBurn" energy-dissipation function, which prevents brake overheating and keeps regenerative braking available on long downhill grades.
- Took part in testing and validation, analysing failures such as the hydrogen inlet pressure valve and defining corrective actions.

3. Graduate Mechanical Engineering Intern, OCP Group Feb 2023 – May 2023

- Led a root cause analysis of critical production-line failures using fishbone diagrams and the 5 Whys.
- Modelled the Heat Recovery System tower in SolidWorks and ran static structural simulations in ANSYS Mechanical to assess external loads.
- Developed a streamlined shell replacement procedure for the tower, which OCP validated and carried out the following year, raising the availability of the line by 15%.

4. Design and Manufacturing Engineering Intern, ENSA Mechanical Laboratory Jun 2022 – Aug 2022

- Reverse-engineered and redesigned an educational torsion test bench in SolidWorks using systematic decomposition and functional analysis.
- Replaced the external comparator with a graduated pulley, improving accuracy and cutting setup time by 80%.
- Manufactured and assembled the parts on conventional and CNC machines.

PROJECTS

1. 3D-Printed 6-DOF Robotic Arm: Design, ROS Control and Dynamics in MATLAB Oct 2024 – 2026

Personal project; dynamics and control work for the Robotic Control Design module (MSc)

- Designed and iterated a 3D-printed arm from 3 to 6 DOF, moving from planetary gearboxes to cycloidal stages on the shoulder and elbow and belt drives on the wrist to reduce backlash. Standardised on NEMA 17 steppers, 608-2Z bearings and parts printable on consumer FDM printers.
- Built a two-layer controller: ROS and MoveIt solve the inverse kinematics for goal poses set in RViz, and a Python layer turns the joint angles into real-time stepper commands.
- Modelled the Lagrangian dynamics of a 3-DOF version in MATLAB (adapted from a simplified PUMA 560 model) with CAD mass properties, linearised it about the worst-case pose into a MIMO transfer-function model, and showed why the linear model misses the velocity-dependent coupling that appears during motion.
- Imported the CAD into Simscape Multibody and tuned three joint PID controllers to track a square path: 0.7 mm peak error at the corners and 0.033 mm in steady state.

MANUSCRIPTS

1. **In preparation:** hardware paper on the planar bipedal robot (mechanical, electrical and operating-software design, calibration and hardware characterisation), for the Journal of Open Hardware. Follow-up papers planned on the ROS 2 software architecture and on trajectory-optimised gait generation.

CERTIFICATION & LANGUAGES

- **Certification:** Fusion 360 Generative Design (Autodesk, via Coursera).
- **Languages:** English (fluent; MSc completed in the UK), French (fluent), Arabic (native).

REFERENCES

1. **Dr Behnam Dadashzadeh**, MSc dissertation supervisor Profile ↗ Scholar ↗
   Lecturer in Mechatronics and Robotics Engineering, Bournemouth University · bdadashzadeh@bournemouth.ac.uk
2. **Dr Roya Haratian, CEng** Profile ↗ ORCID ↗
   Principal Academic in Electronic Science and Engineering, Bournemouth University · rharatian@bournemouth.ac.uk